

Ethosomal Versus Liposomal Vesicular Systems for Topical Curcumin Delivery: A Critical Review of Mechanistic Evidence, Analytical Methodology, and Regulatory Landscape

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Abstract

Background: Curcumin, a naturally occurring polyphenol with well-characterised anti-inflammatory and antioxidant properties, has attracted growing interest as a topical therapeutic for inflammatory dermatoses, wound management, and photoprotection. Two physicochemical challenges limit its clinical development: near-complete aqueous degradation at physiological pH (approximately 90% loss within 30 minutes at pH 7.4, 37°C) and limited passive permeation across the stratum corneum (SC). Vesicular colloidal carriers—conventional phospholipid liposomes and ethanol-enriched ethosomes—have been extensively investigated as delivery platforms to address these barriers.

Scope: This narrative review critically examines the published literature on curcumin-loaded liposomal and ethosomal formulations for topical application. It evaluates comparative physicochemical performance and SC permeation enhancement, provides a mechanistic re-evaluation of confocal laser scanning microscopy (CLSM) data commonly cited in support of intact vesicle penetration, identifies a widespread methodological error in Franz diffusion cell studies involving standard PBS as the receptor phase, reviews the stability analytical requirements specific to ethanol-rich vesicular systems, and maps the current FDA, EMA, and ICH regulatory landscape for topical vesicular drug products.

Key Findings: The literature consistently demonstrates superior SC deposition and permeation enhancement for ethosomes relative to conventional liposomes, attributable primarily to ethanol-mediated fluidisation of SC intercellular lipids rather than intact vesicle transport. CLSM, as routinely applied in skin permeation studies, tracks fluorescent probe distribution in fixed tissue and cannot confirm vesicle integrity during transport—a distinction conflated in a substantial proportion of published ethosome studies. Standard PBS (pH 7.4) is analytically invalid as a Franz diffusion cell receptor phase for curcumin due to near-complete degradation within the experimental timeframe. Headspace GC-FID and Karl Fischer titration are identified as required validated methods for ethanol content and moisture quantification in ethosomal stability programmes. No FDA- or EMA-centrally approved topical liposomal or ethosomal medicinal product currently exists.

Conclusion: This review identifies three correctable gaps in the published ethosome–curcumin literature—mechanistic misinterpretation of CLSM data, invalid receptor phase selection, and inadequate stability characterisation—and provides a framework for addressing each. It further positions topical curcumin vesicular formulations within the applicable regulatory framework, with specific guidance on CMC package requirements.

Keywords: *ethosome; liposome; curcumin; topical drug delivery; stratum corneum; Franz diffusion cell; CLSM; headspace GC-FID; Karl Fischer titration; critical quality attributes; bicyclopentadione; regulatory guidance; skin permeation; phospholipid vesicle*

1. Introduction

Curcumin (1,7-bis(4-hydroxy-3-methoxyphenyl)-1,6-heptadiene-3,5-dione; MW 368.38 g/mol; $\log P \approx 3.29$) is a polyphenolic compound isolated from the rhizome of *Curcuma longa* L. with a substantial evidence base for anti-inflammatory, antioxidant, antimicrobial, and wound-healing activity relevant to dermatological applications [1,2]. Clinical interest in topical curcumin spans inflammatory dermatoses—including psoriasis, atopic eczema, and contact dermatitis—wound management, and photoprotection [4,5]. Despite this pharmacological rationale, no curcumin-containing product has achieved regulatory approval as a topical drug in any major jurisdiction, in part because its physicochemical properties create compounding formulation challenges that the published literature has not always addressed rigorously.

Two principal barriers are well established. First, curcumin's BCS Class IV classification—aqueous solubility below 0.6 $\mu\text{g/mL}$ at neutral pH—means that only a small dissolved fraction is available for percutaneous absorption from any aqueous-based formulation [6]. Second, curcumin undergoes rapid degradation in aqueous environments at near-physiological pH: at pH 7.2–7.4 and 37°C, approximately 90% of dissolved curcumin is lost within 30 minutes through alkaline hydrolysis and subsequent autoxidation of its β -diketone moiety [7,8]. The principal terminal degradation product is bicyclopentadione (BCP), formed via sequential spiroepoxide and vinyl ether intermediates, with secondary products including ferulic acid, vanillin, and acetovanillone [9]. Phosphate-buffered saline (PBS) at pH 7.4 has been identified as a particularly aggressive degradation medium, accelerating autoxidation relative to other buffers at equivalent pH [10].

Phospholipid vesicular carriers—liposomes and ethosomes—represent the most extensively investigated drug delivery strategy for overcoming both the solubility and permeation limitations of curcumin for topical application [11,12]. Conventional liposomes, first described by Bangham, Standish, and Watkins in 1965, are self-assembled bilayer vesicles that can encapsulate both lipophilic and hydrophilic drugs [13]. Applied topically, however, conventional liposomes demonstrate limited penetration beyond the superficial SC layers due to bilayer rigidity [1]. Ethosomes, introduced by Tavitou and colleagues in 2000, incorporate high concentrations of ethanol (20–45% w/w) to produce softer, more deformable vesicles with substantially enhanced SC penetration [14].

Three recurring methodological problems are identifiable across a substantial proportion of published curcumin vesicular delivery studies: (i) mechanistic claims of intact vesicle penetration derived solely from CLSM fluorescence data, which the imaging modality cannot support; (ii) use of standard PBS pH 7.4 as the Franz diffusion cell receptor phase, which causes near-complete curcumin degradation and renders permeation data analytically meaningless; and (iii) inadequate stability methodology for ethanol-rich vesicular systems, specifically the failure to independently quantify ethanol content and moisture by validated methods. This review systematically addresses each of these issues, provides a comparative framework for evaluating liposomal and ethosomal curcumin formulations, and maps the regulatory landscape that any translational development programme must navigate.

2. Curcumin: Physicochemical Profile and Topical Delivery Challenges

2.1 Molecular Properties and BCS Classification

Curcumin exists as three principal tautomers in solution—the bis-keto and two enol forms—with the enol tautomer predominating in organic solvents [6]. Its molecular weight (368.38 g/mol) places it at the upper boundary of what is generally considered acceptable for passive SC permeation under the 500 Da rule of thumb [3], though its high lipophilicity (log P ≈ 3.29) is theoretically favourable for SC partitioning. In practice, log P values above 3 tend to result in effective sequestration in the SC lipid phase, limiting redistribution into viable tissue layers.

The BCS Class IV classification—low aqueous solubility and low membrane permeability—was originally developed to predict oral bioavailability, but the aqueous solubility constraint is directly relevant to topical application: at neutral pH, curcumin solubility is below 0.6 µg/mL, and thermodynamic activity driving passive diffusion from any vehicle is correspondingly low. Complexation, encapsulation, or co-solvent strategies are necessary to deliver therapeutically relevant concentrations to the SC.

2.2 Aqueous Degradation Pathways

Curcumin's chemical instability in aqueous media at physiological pH is multi-mechanistic and has been characterised in detail by several groups. Wang et al. demonstrated pH-dependent first-order degradation kinetics, with a half-life of approximately 8–17 minutes at pH 7.4 and 37°C, increasing dramatically as pH is reduced below 6.5 [8]. Degradation follows two parallel pathways: alkaline hydrolysis of the β-diketone moiety producing ferulic acid as the major product [7], and autoxidation proceeding through a spiroepoxide intermediate to a vinyl ether and ultimately to bicyclopentadione [9]. BCP itself is biologically active, and some anti-inflammatory activities previously attributed to curcumin may be partially due to this degradation product—a complication for pharmacological interpretation [35].

The buffer identity, not merely the pH, also plays a role. Phosphate ions catalyse the autoxidation arm of the degradation pathway, making PBS a particularly aggressive medium relative to HEPES or acetate buffers at equivalent pH [10]. This has direct and critical implications for Franz diffusion cell methodology.

Condition	Approximate t½	Principal Products	Reference
PBS pH 7.4, 37°C	8–17 min	Bicyclopentadione (BCP), ferulic acid, vanillin	[7,8,9]
PBS pH 6.5, 37°C	~2 h	BCP (slower formation)	[8]
PBS pH 5.5, 37°C	>6 h	Minimal degradation	[8,32]
Phospholipid bilayer encapsulation	Substantially extended	Reduced aqueous degradation	[14,15,44]
UV exposure (any pH)	Minutes	Photodegradation products (distinct pathway)	[29,45]

Table 1. Curcumin aqueous degradation under conditions relevant to topical permeation studies (data compiled from cited published literature).

2.3 Implications for Topical Formulation Strategy

The combination of low aqueous solubility, high SC lipid affinity, and rapid aqueous degradation defines the formulation challenge for topical curcumin. An ideal carrier must: (i) solubilise or encapsulate curcumin at therapeutically relevant concentrations; (ii) protect the drug from hydrolytic and oxidative degradation; (iii) enhance partitioning and diffusion across the SC; and (iv) provide a controlled release profile sufficient for local therapeutic effect. Vesicular phospholipid carriers—particularly ethosomes for their combined permeation-enhancement and encapsulation properties—address each of these requirements more completely than simple solutions, gels, or emulsions [14,15].

3. Liposomes and Ethosomes: Comparative Formulation Framework

3.1 Conventional Liposomes

Conventional liposomes are spherical colloidal vesicles formed by the self-assembly of amphiphilic phospholipids—predominantly phosphatidylcholine species—into one or more concentric bilayers surrounding an aqueous core [11]. Their physicochemical characteristics are extensively tuneable: size (50 nm to several micrometres), lamellarity, surface charge, and bilayer fluidity governed by the lipid phase transition temperature [12]. For curcumin encapsulation, the phospholipid bilayer interior provides a lipophilic environment strongly favouring drug partitioning from the aqueous phase [38]. Published entrapment efficiencies for curcumin in conventional liposomes typically range from 60–85% depending on the lipid-to-drug ratio and preparation method [14,18]. The key limitation for topical application is bilayer rigidity: at temperatures below the phospholipid phase transition temperature, the ordered gel-phase bilayer restricts penetration to the superficial SC layers. Cholesterol incorporation, commonly used to improve liposomal stability, further increases bilayer rigidity [12,13].

3.2 Ethosomes

Ethosomes, introduced by Touitou and colleagues in 2000, are distinguished from conventional liposomes by the inclusion of ethanol at concentrations of 20–45% w/w [14]. Ethanol reduces the bilayer bending modulus by intercalating between phospholipid acyl chains, disrupting their ordered packing and creating a more fluid, deformable membrane [15,19]. This fluidisation simultaneously reduces mean vesicle diameter relative to conventional liposomes prepared at equivalent lipid concentrations, increases the accessible lipophilic volume within the bilayer (improving encapsulation of lipophilic drugs such as curcumin), and reduces the energy barrier to vesicle deformation [14,40].

At the SC interface, ethanol exerts a permeation-enhancing effect independent of the vesicle structure itself: it intercalates into ceramide and free fatty acid bilayers of the SC lamellar bodies, transiently disorders their tightly packed alkyl chain organisation, and increases the diffusion coefficients of both drug molecules and the deformable vesicle membrane within the SC intercellular channels [19,20]. The cold ethanol injection method—in which an ethanolic solution of phospholipid and drug is injected into pre-warmed aqueous phase under stirring—is the most widely used preparation approach, yielding vesicles in the 100–250 nm size range [14].

Property	Conventional Liposomes	Ethosomes	Mechanistic Basis
Ethanol content (% w/w)	0–5	20–45	Bilayer fluidiser; critical excipient [14]
Typical Z-average diameter (nm)	100–200	100–250	Ethanol reduces bending modulus [14,15]
Bilayer rigidity	High (rigid)	Low (deformable)	Ethanol disrupts acyl chain packing [15,19]
SC penetration depth	Superficial SC only	SC + viable epidermis	Ethanol-mediated lipid fluidisation [16,20]
Encapsulation efficiency (lipophilic drugs)	Moderate	Higher	Expanded lipophilic bilayer volume [15,40]
Primary stability CQAs	Diameter, PDI, EE%, phospholipid oxidation	Diameter, PDI, EE%, ethanol content, moisture	Ethanol loss → rigidity; moisture → aggregation [23,28]

Table 2. Comparative physicochemical properties of conventional liposomes and ethosomes relevant to topical curcumin delivery.

3.3 Critical Quality Attributes

For both vesicular systems, hydrodynamic diameter (Z-average), polydispersity index (PDI), zeta potential, and entrapment efficiency (EE%) are the primary characterisation CQAs. For curcumin specifically, EE% should be determined by ultracentrifugation rather than dialysis or gel filtration, to avoid re-equilibration artefacts from the drug's high lipophilicity [39].

For ethosomal formulations specifically, ethanol content and free water content emerge as unique and primary CQAs. Ethanol content directly governs bilayer deformability and hence SC permeation performance: published data indicate that concentrations below approximately 35% w/w are associated with a shift toward conventional liposome-like rigidity, substantially diminishing the permeation advantage [40]. Moisture content is a CQA because free water promotes Ostwald ripening, vesicle fusion, and aggregation [28]. These two CQAs cannot be independently quantified by loss-on-drying; headspace GC-FID and Karl Fischer titration are the required methods.

4. Critical Methodological Framework

4.1 Mechanistic Interpretation of CLSM Data

Confocal laser scanning microscopy of fluorescent probe-loaded vesicles in skin cross-sections has become the standard visualisation tool in the ethosome literature. The observation of greater fluorescence depth in ethosome-treated sections relative to liposome-treated sections has been widely cited as evidence of intact vesicle penetration into the SC. This interpretation is mechanistically unsupported and requires correction.

CLSM of fixed, cryosectioned skin tissue images the spatial distribution of fluorescent molecules at the time of fixation. It cannot distinguish between fluorescence from intact vesicle-associated probe, probe molecules that have dissociated from the vesicle membrane and diffused freely through SC intercellular channels, or probe deposited within SC corneocytes [16,17]. Two

landmark studies established this limitation clearly. Kirjavainen et al. demonstrated that lipophilic fluorescent probes dissociate from liposomal membranes within the SC environment, generating CLSM distribution patterns indistinguishable from free probe controls [17]. Verma and Fahr showed that ethanol and phospholipid act cooperatively to enhance topical delivery through a bilayer-fluidisation mechanism, with probe distribution patterns consistent with ethanol-mediated SC lipid fluidisation rather than intact carrier permeation [16].

The mechanistic consensus supported by biophysical evidence is that ethanol—at the concentrations employed in ethosomes—intercalates into SC ceramide and free fatty acid bilayers, transiently reduces the diffusion activation energy within the SC intercellular matrix, and enables both free drug molecules and the deformable ethosomal membrane to penetrate more deeply than rigid conventional liposomes [15,19,20]. Investigators reporting enhanced fluorescence depth in CLSM studies as 'evidence of intact vesicle penetration' should revise this interpretation to 'evidence of enhanced probe molecule penetration consistent with ethanol-mediated SC fluidisation.'

4.2 Franz Diffusion Cell Receptor Phase Selection for Curcumin

A critical and pervasive methodological error in the published curcumin topical permeation literature is the use of phosphate-buffered saline at pH 7.4 as the Franz diffusion cell receptor phase. This practice is analytically invalid for curcumin and produces data that are not interpretable as permeation measurements.

Curcumin has a degradation half-life of 8–17 minutes in PBS at pH 7.4 and 37°C [7,8]. In a standard 24-hour Franz diffusion cell experiment, the vast majority of any curcumin that permeates the skin and enters the PBS receptor phase will have degraded before sampling. Detected UV absorbance or fluorescence at curcumin wavelengths reflects the combined signal of intact curcumin (a minor fraction), BCP, ferulic acid, and other degradation products—not curcumin permeation per se.

A methodologically valid receptor phase for curcumin Franz diffusion studies requires: (i) a pH sufficiently low to substantially reduce degradation kinetics — pH 5.0–5.5 is appropriate and within the physiologically relevant range for the skin surface [8,32]; (ii) a solubilising agent to maintain sink conditions — hydroxypropyl- β -cyclodextrin (HP β CD) at 0.5% w/v increases apparent solubility and confers additional chemical stabilisation [32]; (iii) an antioxidant to suppress the autoxidative degradation pathway — L-ascorbic acid at 0.1% w/v is effective [34]; and (iv) oxygen exclusion through de-aeration and continuous nitrogen sparging [9]. The modified receiver should be validated to demonstrate $\geq 85\%$ curcumin recovery over the full experimental duration at 37°C. LC-MS/MS confirmation of receptor phase samples using MRM transitions for intact curcumin (m/z 367 $\rightarrow$ 177, [M-H]⁺) and BCP (m/z 273 $\rightarrow$ 229) allows accurate mass balance assessment and distinguishes permeated intact drug from in-situ degradation products [9,35].

4.3 Stability Methodology for Ethosomal Formulations

An ICH Q1A(R2)-compliant stability programme for ethosomal formulations must monitor the CQAs described in Section 3.3 under standard stability conditions—4°C, 25°C/60% RH, and 40°C/75% RH—with analysis at T = 0, 1, 3, and 6 months as a minimum.

Ethanol content quantification requires static headspace GC-FID, validated per ICH Q2(R1), using a DB-WAX or equivalent polar capillary column, an appropriate internal standard (n-propanol at 0.10% v/v), and a five-point external calibration spanning the relevant range. Although ethanol is classified as an ICH Q3C Class 3 residual solvent [27], in ethosomal formulations it functions as a primary functional excipient and should be treated as a formulation CQA with a justified lower specification limit—typically $\geq 35\text{--}38\%$ w/w, based on the demonstrated concentration-dependent relationship between ethanol content and bilayer deformability [40].

Water content must be determined by Karl Fischer titration per Ph. Eur. 2.5.12 and USP **921** [28] rather than loss-on-drying. Loss-on-drying reports the combined volatile fraction of ethanol and water together and cannot distinguish between them. Ethanol depletion combined with water ingress during storage is the most important degradation pathway for ethosomal physical integrity: ethanol loss increases bilayer rigidity toward that of a conventional liposome, while water ingress drives osmotic vesicle swelling and aggregation [28,36,40]. Both effects manifest as increases in Z-average diameter and PDI on stability testing.

5. Comparative In Vitro Skin Permeation: Review of Published Evidence

5.1 Published Comparative Studies

The published literature consistently reports superior SC permeation and drug retention for curcumin-loaded ethosomes relative to conventional liposomes, though the magnitude of enhancement varies substantially across studies, reflecting differences in formulation composition, skin model, receptor phase, and analytical methodology. Enhancement ratios for cumulative permeation of 2–5-fold are commonly reported, with SC retention enhancements in the range of 1.5–3-fold [14,15,18]. Porcine ear skin, accepted under OECD TG 428 [21], is the most widely used surrogate for human skin in in vitro permeation studies.

5.2 Methodological Confounds in the Published Literature

Several systematic confounds reduce the reliability of the cumulative published data on curcumin vesicular permeation. The receptor phase issue discussed in Section 4.2 is the most consequential: studies using PBS pH 7.4 as the receiver cannot have measured curcumin permeation quantitatively. The CLSM interpretation issue discussed in Section 4.1 similarly affects a large proportion of mechanistic claims in the literature.

A third confound specific to comparative studies is the equimolar drug loading assumption: ethosomes and liposomes often differ substantially in EE%, meaning that equal nominal drug concentration in the donor formulation may correspond to very different encapsulated drug fractions. Permeation comparisons are only interpretable when the encapsulated drug fraction is matched, or when both encapsulated and free drug are quantified separately at T = 0.

A fourth consideration is the role of the ethanolic solution control. Many comparative studies test ethosomes versus liposomes without including a matched ethanol-in-water solution at equivalent ethanol concentration as a control arm. This control is essential to isolate the phospholipid vesicle contribution from the ethanol permeation-enhancement contribution. Where this control has been included, the phospholipid component typically contributes an additional 1.5–2.0-fold enhancement over the matched ethanolic solution alone [16], consistent with the cooperative mechanism described by Verma and Fahr.

6. Regulatory Landscape for Topical Vesicular Drug Products

6.1 Current Approval Status

No FDA- or EMA-centrally approved topical (cutaneous) liposomal or ethosomal medicinal product currently exists [47,48,52]. All FDA-approved liposomal drug products are parenteral or pulmonary in route of administration—including liposomal doxorubicin (Doxil®), liposomal amphotericin B (AmBisome®), liposomal irinotecan (Onivyde®), and liposomal bupivacaine (Exparel®) [47,48]—and the regulatory pathways, quality expectations, and bioequivalence frameworks developed for these products are not directly transferable to topical applications. This regulatory gap means that any development programme for a topical liposomal or ethosomal drug product must construct its regulatory strategy without precedent approval as a reference point.

Regulatory Body	Key Guidance	Relevance	Gap / Challenge
US FDA	Liposome Drug Products CMC Guidance (2018)	CQA expectations: size, lamellarity, EE%, in vitro release	No approved topical liposomal product; IVRT/DPK not extended to vesicular systems [47,50]
EMA	Guideline on Quality and Equivalence of Topical Products (2019)	Physicochemical differences in vehicle may require clinical BE data	No centrally authorised topical liposomal/ethosomal product [52]
ICH	Q1A(R2), Q1B, Q2(R1), Q3C, Q6A, Q8(R2)	Stability, photostability, analytical validation, QbD framework	Guidelines not vesicular-specific; sponsor must adapt
Indian CDSCO	New Drugs and Clinical Trials Rules 2019	IND/CTA required for new topical drug; AYUSH framework for botanical	No specific vesicular topical guidance published

Table 3. Regulatory framework summary for topical vesicular drug product development.

6.2 CMC Requirements for Topical Curcumin Vesicular Development

Curcumin is not an approved active pharmaceutical ingredient for any indication in the US, EU, or UK. Development of a curcumin-based topical drug product therefore requires full IND-enabling studies (US) or a Clinical Trial Application with a complete investigational medicinal product dossier (EU). The CMC package must address API characterisation including curcuminoid purity profile (HPLC-UV/LC-MS/MS), polymorphism, and particle size distribution; formulation development rationale under ICH Q8(R2) QbD principles [41]; and product specifications per ICH Q6A [42] referencing all relevant CQAs.

For an ethosomal topical formulation specifically, the CMC package should document: Z-average diameter and PDI with justified acceptance criteria; zeta potential; EE% by ultracentrifugation; in vitro drug release using a validated modified receptor phase; ethanol content (GC-FID; lower specification limit justified by deformability data); moisture content (Karl Fischer); vesicle integrity at defined stability time points; photostability per ICH Q1B [45]; and comparative in vitro performance versus a non-vesicular control. This documentation aligns with the FDA liposome guidance [46], the EMA topical equivalence guideline [51], and the applicable ICH quality guidelines.

7. Future Perspectives

Several directions offer meaningful opportunities to advance the field of topical curcumin vesicular delivery. First, the mechanistic question of whether the deformable ethosomal membrane traverses SC intercellular channels as an intact vesicle remains genuinely unresolved. Cryo-TEM of rapidly frozen, unfixed skin cross-sections, combined with correlative fluorescence microscopy using probes of defined log P values, would provide orthogonal evidence that CLSM alone cannot supply.

Second, the receptor phase problem identified in this review—the analytical invalidity of PBS pH 7.4 for curcumin Franz diffusion studies—extends to many other phenolic and polyphenolic compounds with similar pH-dependent instability. A systematic methodological survey establishing appropriate receptor phase conditions for the broader class of bioactive polyphenols under investigation for topical delivery would have considerable practical value.

Third, scale-up and manufacturing consistency for ethosomal formulations remain underexplored. The cold ethanol injection method is well established at laboratory scale, but maintaining ethanol content within a narrow specification during continuous or large-batch manufacture, and demonstrating CQA consistency across batches, are process validation challenges without published regulatory precedent. Process analytical technology tools—online DLS for particle size monitoring, in-line GC for ethanol content—may offer pathways to real-time release testing for future ethosomal drug products.

8. Conclusions

Ethosomes consistently outperform conventional liposomes for SC permeation and drug retention of curcumin across the published comparative literature. The primary mechanism is ethanol-mediated fluidisation of SC intercellular lipids, not intact vesicle transport. The phospholipid vesicle component contributes an additional permeation enhancement beyond ethanol alone, consistent with a cooperative ethanol-phospholipid mechanism.

CLSM fluorescence depth profiles—the most commonly cited evidence for intact ethosome penetration—track probe molecule distribution in fixed tissue sections and cannot confirm vesicle integrity during transport. This methodological limitation has been conflated with mechanistic evidence of intact vesicle penetration in a substantial proportion of the published literature and should be corrected in future reporting.

Standard PBS at pH 7.4 is analytically invalid as a Franz diffusion cell receptor phase for curcumin, with near-complete drug degradation occurring within the experimental timeframe. A modified receiver—PBS pH 5.5, HP β CD, ascorbic acid, nitrogen sparging—validated at $\geq 85\%$ curcumin recovery over 24 hours is the minimum methodological requirement.

Headspace GC-FID and Karl Fischer titration are the required analytical methods for independent quantification of ethanol and moisture in ethosomal stability programmes. Loss-on-drying is unsuitable for this application. No FDA- or EMA-centrally approved topical liposomal or ethosomal medicinal product currently exists; development programmes must proceed without approved precedent, drawing on FDA liposome CMC guidance, the EMA topical equivalence guideline, and the applicable ICH quality guidelines.

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